

Affine extension from equal-radius closed balls

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_closedBall`

Extends an isometry of closed balls by first recovering the centers and then passing through the convex-interior extension theorem.

Statement

Let f be a surjective isometry between the closed balls $\text{closedBall}(c,r)$ and $\text{closedBall}(d,r)$ of the same radius $r > 0$ in real normed spaces. Then f has a unique ambient real affine isometry-equivalence extension agreeing with f on the whole source closed ball.

Assumptions

Real normed vector spaces; f is an isometry equivalence between closed balls with the same radius $r > 0$.

Conclusion

f extends uniquely to an ambient real affine isometry equivalence.

Proof idea

The center-transport theorem shows that f sends c to d because both closed balls are bounded and invariant under reflection about their centers. Distance preservation then identifies membership in the two ambient interiors, which are the corresponding open balls. The convex extension theorem completes the proof.

Proof steps

1. Regard c and d as points of their respective closed balls.
2. Use boundedness and point-reflection invariance of closed balls to prove that f maps the source center to the target center.
3. For each source point, rewrite interior membership as a strict distance-to-center inequality and transport that distance through f using the center identity.
4. Show that the source closed ball has nonempty ambient interior because r is positive.
5. Apply the convex-set extension theorem to the closed ball, the nonempty interior, and the proved interior-membership equivalence.

Main citations

`MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

The proof or construction of Affine extension from equal-radius closed balls uses the project declaration “Center transport under bounded point-reflection symmetry” at the indicated step.

`_private.MathlibAnnex.Analysis.Normed.Affine.IsometryExtension.Ball.0.MathlibAnnex.IsometryEquiv.mapsTo_pointReflection_closedBall`

The proof or construction of Affine extension from equal-radius closed balls uses the project declaration “Closed balls are invariant under center reflection” at the indicated step.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_of_convex`

The proof or construction of Affine extension from equal-radius closed balls uses the project declaration “Convex-set extension via ambient interiors” at the indicated step.

Lean source signature (exact)

```
theorem existsUnique_affineExtension_closedBall
{c : E} {d : F} {r : ℝ} (f : closedBall c r ≈i closedBall d r)
(hr : 0 < r) :
∃! A : E ≈a i[ℝ] F,
∀ x : closedBall c r,
A (x : E) = ((f x : closedBall d r) : F)
```

[Read exact MathlibAnnex v0.4.0 source](#)

Exact Card identity

Stable Card ID: 92f93cdb5b6da9a1e1a1837b4e8470509909095a54e82328e641ff0b3c842995

Card SHA-256: 8f8f74883b3320a2f4b6f780cbbf954685b42419f699eb7498b3a5f62286e6f5

Approved exposition SHA-256: 2f7bf290b978504895210d7d80094d4e2d1e83f0f45a7887e54ae6ff0e184f55

AI-assisted material. Selected, edited, and reviewed by Ryotaro Tanaka. The recorded review is source–exposition correspondence, not an independent proof audit.